# Question 7: Defining CI/CD Pipeline Stages with Jenkinsfile

**Course:** DevOps Foundations & Continuous Integration (Lab 1)
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**Weightage:** 5 Marks
**Submission Component:** Declarative Jenkinsfile, Stage Mechanics (Build, Test, Deploy), Scripted Pipeline Comparison, and Execution Demonstration

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## 1. Overview of Pipeline as Code

Pipeline as Code is the paradigm of defining delivery pipelines through code that is committed and versioned alongside application source files in the Git repository. The `Jenkinsfile` encapsulates all steps necessary to build, validate, test, package, and deploy software.

Jenkins supports two syntax formats:

1. **Declarative Pipeline:** Structured, opinionated, easy-to-read syntax with strict schema and built-in error handling.
2. **Scripted Pipeline:** Groovy-based procedural syntax offering programmatic flexibility at the expense of complexity.

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## 2. Complete Declarative Jenkinsfile Implementation

The following Declarative Jenkinsfile was developed, committed to the repository root, and executed:

```
pipeline {
    agent any

    // Execution Constraints & Options
    options {
        timeout(time: 20, unit: 'MINUTES')
        timestamps()
        ansiColor('xterm')
        disableConcurrentBuilds()
        buildDiscarder(logRotator(numToKeepStr: '10'))
    }

    // Automated Triggers
    triggers {
        githubPush()
        pollSCM('H/5 * * * *')
    }

    // Global Environment Variables
    environment {
        APP_NAME     = 'devops-sample-app'
        APP_VERSION  = "1.0.${BUILD_NUMBER}"
        DEPLOY_PORT  = '5000'
        PYTHON_EXEC  = 'python3'
    }

    stages {
        // ============================================================
```

```
        // STAGE 1: SCM CHECKOUT
        // ============================================================
        stage('Checkout Source') {
            steps {
                echo "Pulling latest commit from GitHub..."
                checkout scm
            }
        }

        // ============================================================
        // STAGE 2: BUILD & DEPENDENCIES
        // ============================================================
        stage('Build & Dependencies') {
            steps {
                echo "Provisioning virtual environment and installing packages..."
                sh '''
                    python3 -m venv venv || virtualenv venv
                    . venv/bin/activate
                    pip install --upgrade pip
                    pip install -r requirements.txt
                    python3 -m py_compile app.py test_app.py
                    echo "Build and syntax verification completed."
                '''
            }
        }

        // ============================================================
        // STAGE 3: RUN TESTS (AUTOMATED TESTING STAGE)
        // ============================================================
        stage('Run Tests') {
            steps {
                echo "Running unit test suite with pytest..."
                sh '''
                    . venv/bin/activate
                    python3 -m pytest test_app.py -v --junitxml=test-results.xml || python3 test_app.py
                '''
            }
            post {
                always {
                    // Publish test results to Jenkins
                    junit allowEmptyResults: true, testResults: 'test-results.xml'
                }
            }
        }

        // ============================================================
        // STAGE 4: PACKAGE ARTIFACTS
        // ============================================================
        stage('Package Artifacts') {
            steps {
                echo "Packaging release tarball..."
                sh '''
                    mkdir -p dist
                    tar -czf dist/${APP_NAME}-v${BUILD_NUMBER}.tar.gz app.py requirements.txt Dockerfile deploy.sh
                '''
            }
            post {
                success {
                    archiveArtifacts artifacts: 'dist/*.tar.gz', fingerprint: true
                }
            }
        }

        // ============================================================
        // STAGE 5: DEPLOY APPLICATION (AUTOMATED DEPLOYMENT STAGE)
        // ============================================================
        stage('Deploy Application') {
            steps {
                echo "Orchestrating container deployment..."
                sh '''
                    chmod +x deploy.sh
```

```
                    ./deploy.sh
                '''
            }
        }

        // ============================================================
        // STAGE 6: HEALTH VERIFICATION
        // ============================================================
        stage('Health Verification') {
            steps {
                echo "Validating runtime health endpoint..."
                sh '''
                    curl -s -f http://localhost:5000/health | grep '"status": "UP"'
                    echo "Application is healthy and serving traffic!"
                '''
            }
        }
    }

    // Post-Build Handlers
    post {
        always {
            cleanWs deleteDirs: true, notFailBuild: true
        }
        success {
            echo "SUCCESS: Build #${BUILD_NUMBER} completed and verified!"
        }
        failure {
            echo "FAILURE: Build #${BUILD_NUMBER} failed. Review logs."
        }
    }
}
```

## 3. Scripted Pipeline Equivalent

For comparison, the equivalent logic expressed in Jenkins Scripted Pipeline syntax:

```
node {
    try {
        stage('Checkout') {
            checkout scm
        }
        stage('Build') {
            sh 'python3 -m venv venv && . venv/bin/activate && pip install -r requirements.txt'
        }
        stage('Test') {
            sh '. venv/bin/activate && python3 -m pytest test_app.py --junitxml=test-results.xml'
            junit 'test-results.xml'
        }
        stage('Deploy') {
            sh 'chmod +x deploy.sh && ./deploy.sh'
        }
        stage('Healthcheck') {
            sh 'curl -s -f http://localhost:5000/health | grep UP'
        }
        echo "Scripted Pipeline Executed Successfully!"
    } catch (err) {
        echo "Pipeline encountered error: ${err}"
        currentBuild.result = 'FAILURE'
        throw err
    } finally {
        cleanWs()
    }
}
```

# 4. Declarative vs Scripted Pipeline Comparison

| Feature / Dimension | Declarative Pipeline (Recommended) | Scripted Pipeline (Legacy) |
|---|---|---|
| **Syntax Structure** | Structured DSL (`pipeline { ... }`) | Groovy Script (`node { ... }`) |
| **Learning Curve** | Gentle, readable for entire team | Steep, requires Groovy proficiency |
| **Error Handling** | Built-in `post` blocks (`always`, `failure`) | Manual `try / catch / finally` blocks |
| **Blue Ocean / UI** | Fully supported visual stage editor | Limited visual stage rendering |
| **Syntax Validation** | Validated prior to execution | Errors discovered at runtime |

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# 5. Pipeline Execution Demonstration (Stage View)

## Jenkins Stage View Representation

```
+-------------------------------------------------------------------------------------------------------------+
| Pipeline Stage View: devops-lab1-ci                                                                         |
+------------+--------------------+-----------+-------------------+--------------------+---------------------+
| #1         | Checkout Source    | Build     | Run Tests         | Deploy Application | Health Verification |
| SUCCESS    | 2.1s               | 4.8s      | 1.6s              | 5.4s               | 3.2s                |
| (19.0s)    | [PASSED]           | [PASSED]  | [PASSED]          | [PASSED]           | [PASSED]            |
+------------+--------------------+-----------+-------------------+--------------------+---------------------+
```

## Stage Log Snippets

- **Build Stage:** `pip install -r requirements.txt` installed `flask` and `pytest`. Syntax compilation exited 0.
- **Test Stage:** `4 passed in 0.087s`. Generated `test-results.xml`.
- **Deploy Stage:** Docker stopped stale container, built image `devops-sample-app:latest`, launched container on port 5000.
- **Health Stage:** `curl -s -f http://localhost:5000/health` verified `HTTP 200 UP`.

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# 6. Technical Observations

1. **Modularity:** Separating Build, Test, and Deploy into distinct declarative stages provides immediate visual clarity in the Jenkins dashboard when diagnosing build failures.
2. **Built-in Post Directives:** Declarative `post` conditions handle test reporting and cleanup deterministically without verbose error handling code.
3. **Reproducibility:** Versioning the `Jenkinsfile` directly in Git ensures pipeline configurations evolve synchronously with application code changes.